

Eigenspaces of a Rank-One Matrix

Topic.

Linear algebra, rank-one matrices, eigenvalues, eigenspaces.

Statement.

Let

$$a_1, \dots, a_n, \quad b_1, \dots, b_n$$

be nonzero numbers, and let

$$A = \begin{pmatrix} a_1 b_1 & a_1 b_2 & \cdots & a_1 b_n \\ a_2 b_1 & a_2 b_2 & \cdots & a_2 b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n b_1 & a_n b_2 & \cdots & a_n b_n \end{pmatrix}.$$

- (a) Find the dimension of the eigenspace corresponding to the eigenvalue $\lambda = 0$.
- (b) Prove that A has at most one nonzero eigenvalue. Also prove that if $\mu \neq 0$ is an eigenvalue of A , then the dimension of the corresponding eigenspace is 1.

Solution.

(a)

Let

$$\lambda = 0.$$

Then

$$AX = 0, \quad X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Write the product explicitly. The first row gives

$$a_1 b_1 x_1 + a_1 b_2 x_2 + \cdots + a_1 b_n x_n = a_1 (b_1 x_1 + \cdots + b_n x_n).$$

The second row gives

$$a_2 (b_1 x_1 + \cdots + b_n x_n),$$

and so on.

Therefore,

$$AX = 0$$

is equivalent to the system

$$\begin{cases} a_1(b_1x_1 + \cdots + b_nx_n) = 0, \\ a_2(b_1x_1 + \cdots + b_nx_n) = 0, \\ \cdots \\ a_n(b_1x_1 + \cdots + b_nx_n) = 0. \end{cases}$$

Since

$$a_i \neq 0, \quad \forall i = 1, \dots, n,$$

this is equivalent to

$$b_1x_1 + \cdots + b_nx_n = 0.$$

Since

$$b_n \neq 0,$$

we can express

$$x_n = -\frac{b_1}{b_n}x_1 - \frac{b_2}{b_n}x_2 - \cdots - \frac{b_{n-1}}{b_n}x_{n-1}.$$

Hence the general solution has the form

$$X = x_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ -\frac{b_1}{b_n} \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ -\frac{b_2}{b_n} \end{pmatrix} + \cdots + x_{n-1} \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ -\frac{b_{n-1}}{b_n} \end{pmatrix}.$$

Thus the eigenspace for $\lambda = 0$ has dimension

$$\dim E_0 = n - 1.$$

(b)

Let

$$\lambda \neq 0$$

be an eigenvalue, and let

$$X \neq 0$$

be an eigenvector. Then

$$AX = \lambda X.$$

Put

$$S = b_1x_1 + \cdots + b_nx_n.$$

By the same computation as above,

$$AX = \begin{pmatrix} a_1S \\ a_2S \\ \vdots \\ a_nS \end{pmatrix}.$$

From

$$AX = \lambda X,$$

we get

$$\begin{cases} a_1S = \lambda x_1, \\ a_2S = \lambda x_2, \\ \cdots \\ a_nS = \lambda x_n. \end{cases}$$

Since

$$\lambda \neq 0,$$

we have

$$x_i = \frac{S}{\lambda}a_i, \quad \forall i = 1, \dots, n.$$

Therefore,

$$X = \frac{S}{\lambda} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}.$$

So every eigenvector corresponding to a nonzero eigenvalue is proportional to

$$a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}.$$

Hence if

$$\lambda \neq 0$$

is an eigenvalue, then its eigenspace consists of vectors of the form

$$t \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}, \quad t \neq 0.$$

Therefore,

$$\dim E_\lambda = 1.$$

Now prove that nonzero eigenvalues are not more than one.

Suppose there are two nonzero eigenvalues

$$\lambda_1 \neq \lambda_2.$$

We have already shown that every eigenvector corresponding to a nonzero eigenvalue is proportional to the vector

$$a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}.$$

Thus the same nonzero vector a must be an eigenvector for both λ_1 and λ_2 :

$$Aa = \lambda_1 a, \quad Aa = \lambda_2 a.$$

Hence

$$\lambda_1 a = \lambda_2 a.$$

Since

$$a \neq 0,$$

we get

$$\lambda_1 = \lambda_2.$$

Contradiction.

Therefore, A has at most one nonzero eigenvalue.